What an amplitude-estimation error bar actually controls

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Error budget and quadrature

A certified estimate of a badly encoded amplitude is not a certified price or CVaR. Linking my regime-conditional factor models to amplitude estimation, I separate target, encoding and estimation errors; no novelty, new pricing method or interval-arithmetic proof is claimed. For $S_0 = 100$, $K = 105$, $r = 0.05$, $\sigma = 0.2$, $T = 1$ year, and standard-normal distribution/density Φ, ϕ, the call and normal-loss references are

$$C = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2) = 8.0213522351, \qquad d_{1,2} = \frac{\log(S_0/K) + (r \pm \sigma^2/2)T}{\sigma\sqrt{T}},$$

$$L \sim N(\mu, \sigma_L^2): \quad q_\alpha = \mu + \sigma_L\Phi^{-1}(\alpha), \qquad \mathrm{CVaR}_\alpha(L) = \mu + \frac{\sigma_L\phi(\Phi^{-1}(\alpha))}{1-\alpha}.$$

At $\mu = 0$, $\sigma_L = 1$, $\alpha = 0.95$: VaR 1.6448536270, CVaR 2.0627128075 (standard-normal loss units; `problems.py`, `test_audit.py`, fresh `run.py` output in `verification/run_report.txt`). For masses p_i, $0 \le f_i \le 1$, $a = \sum_i p_i f_i$, canonical estimation returns $\hat{a} = \sin^2(\pi k/t)$, $t = 2^m$ phase bins. In `estimation.py`, Brassard-Hoyer-Mosca-Tapp gives

$$\Pr\{|\hat{a} - a| \le B_t(a)\} \ge \frac{8}{\pi^2} = 0.810569\ldots, \qquad B_t(a) = \frac{2\pi\sqrt{a(1-a)}}{t} + \frac{\pi^2}{t^2}.$$

This ideal-oracle, probabilistic a priori bar uses known encoded a; otherwise $a(1-a) \le 1/4$ gives $B_t(a) \le \pi/t + \pi^2/t^2$. It is not a shot-derived interval; scaling converts units, not coverage of quadrature, quantile, sampling or hardware error. At fixed encoding/success probability, ideal query improvement is quadratic ($1/\epsilon$ versus Monte Carlo's $1/\epsilon^2$), not exponential: $t-1$ uses of Q, plus loading/compilation costs. Statevector simulation is not a wall-clock speedup. With discounted payoff maximum s, $C_h = sa_h$, $\widehat{C_h} = s\hat{a}_h$:

$$\widehat{C_h} - C = (\widehat{C_h} - C_h) + (C_h - C), \quad C_h - C = (C_h - C_W) + (C_W - C).$$

Only estimation has bar $sB_t(a_h)$; price certification also needs encoding error, measured here against a closed form without interval enclosure. Uniform **spot** nodes spanning the range corresponding to shocks $[-4, 4]$ use normalized lognormal-density masses: $C_W = 8.0177885991$, bias $C_W - C = -0.0035636360$. The $\pm 4\sigma$ window sets an asymptotic floor, not a finite-grid lower bound: cancellation gives 0.00061609-dollar error at 96 nodes; at 256, price 8.0178887418. Full-domain Gauss-Hermite is non-monotone on the strike kink despite polynomial exactness, and worse at 16 nodes. One small error proves neither convergence nor a general quadrature ranking.

Nodes	Gauss-Hermite price	Signed error	Uniform-grid price	Signed error
16	8.19192793	+0.17057570	8.00851991	-0.01283233
64	8.03695667	+0.01560444	8.01769175	-0.00366048
96	8.02005137	-0.00130087	8.02073615	-0.00061609
128	8.01162193	-0.00973030	8.01852537	-0.00282686
256	7.99900672	-0.02234551	8.01788874	-0.00346349
512	8.01974650	-0.00160574	8.01775543	-0.00359680

Source: `data/scale_sweep.json`, `classical`; prices/signed errors in dollars versus Black-Scholes.

Figure 1 (`figures/scaling_methods.png`): `data/scale_sweep.json`, regenerated 23 October 2026; noisy panel: unmitigated 15 April 2026 FakeKingston/Aer simulation. Monte Carlo uses 24 seeds/path count and separately simulated payoff SD for its dashed standard-error reference, not a rigorous finite-sample bound; node/path/phase-bin budgets imply no equal end-to-end costs.

Amplitude estimation against its bar

For $a = \sin^2(0.7) = 0.4150164285$, medians are descriptive, classically computed probability-weighted medians of the exact statevector distribution, not shots or experimental repetitions. Success probabilities describe random

outcomes: they may decrease as bars narrow; stationary phase-bin medians are not failures. The minimum remains above $8/\pi^2$.

Phase bins t	Absolute median error	Amplitude bar	Probability inside bar
16	0.08498357	0.23204574	0.828128
64	0.01256159	0.05078272	0.894985
128	0.01161833	0.02478897	0.811465
256	0.00049737	0.01224389	0.996277
1024	0.00049737	0.00303273	0.949507

The 16-node call uses $a_h = 0.0677149409$, $s = 118.268137$ dollars, and grid reference 8.0085199051, not Black-Scholes. At $t = 16$ its bar exceeds the price; at $t = 1024$ the Black-Scholes error +0.022486 dollars combines positive estimation error and negative grid bias.

t	Median price	Error vs grid	Price bar	Probability inside
16	4.501313	-3.507207	16.228894	0.921119
64	6.982476	-1.026044	3.202297	0.843049
256	7.682285	-0.326235	0.747141	0.814712
1024	8.043838	+0.035319	0.183446	0.933545

Both tables: `data/scale_sweep.json`, `amplitude_estimation`; call prices, signed errors and bars in dollars.

Known-amplitude/call seeded shots satisfy their bars; Gaussian CVaR retains its miss. For 20,000 losses, seed 11, 32 bins: excess assembly 2.0603234871; $t = 16$ median 3.2189159024, bar 3.245422, **seed-0 single shot outside its bar**, success probability 0.858922, consistent with the theorem. At $t = 1024$: median 2.0606839698, bar 0.02604439 (`run.py`, existing audit). No filtering or every-shot guarantee.

Figure 2 (`figures/scaling_details.png`): `data/scale_sweep.json`, regenerated 23 October 2026; shots/legacy baseline retained, no-CZ panel: unmitigated 15 April 2026 FakeKingston/Aer simulation.

Simulated noise

FakeKingston is the ibm_kingston calibration snapshot of 15 April 2026 in Qiskit Aer, not live hardware; no mitigation. This sweep (earlier runs: `MEASUREMENTS.md`) averages eight seeds, 4,096 shots each, without call-loader amplification. Shift from the grid grows with CZ count, alongside changing nodes/depth/compilation: no isolated CZ causality. More shots reduce fluctuations, not bias.

Spot nodes	CZ gates	Grid price	Ideal Aer mean	Noisy Aer mean	Noisy minus grid
4	7	4.017360	3.981011	5.688189	+1.670829
8	23	8.037867	8.034206	10.585951	+2.548085
16	62	8.008520	7.860962	12.816472	+4.807952
32	132	7.983626	7.850134	17.761154	+9.777529

Source: `data/scale_sweep.json`; prices and shifts in dollars.

The rough CZ-survival check mixes with 1/2: $w = (1-p)^g$, $a_{\mathrm{mix}} = wa + (1-w)/2$. For $p = 0.0016$, $g = 62$, $w \simeq 0.9055$, $a = 0.0677$: prediction $a_{\mathrm{mix}} \simeq 0.109$, measured 0.108368. Omitting relaxation/readout/individual-gate details, this is no fit, mechanism proof, bound or price adjustment. The one-qubit no-CZ control follows $\sin^2((2m+1)\theta)$, $\theta = 0.7$: at $m = 4$, ideal 0.00028271, noisy 0.01234436 (floor near a zero). Averaging 16 seeds, 4,096 shots each, largely preserves oscillations through $m = 16$, without testing multiqubit preparation. Earlier canonical $t = 4, 8$ runs: ideal/noisy medians 1/2, error 0.08498, ideal bars $1.391, 0.541$. Wide bars diagnose little; coarse medians hide distribution changes; the ideal theorem does not certify noise.

Established work: Montanaro, *Quantum speedup of Monte Carlo methods* (2015); Woerner and Egger, *Quantum risk analysis* (2019); Tanaka et al., *Amplitude estimation via maximum likelihood on noisy quantum computer* (2021): prior work, no novelty claim.

Empirical CVaR, reproduction and an open question

For loss = minus monthly return and $\alpha = 0.95$, distinguish sample tail mean S, legacy comparator-selected histogram mean H, and excess assembly D. With sample quantile q, centers c_i, masses p_i:

$$D = q + \frac{\sum_i p_i (c_i - q)_+}{1-\alpha}, \quad \widehat{D} = q + \frac{E_{\max}}{1-\alpha}\hat{a}, \quad E_{\max} = \max_i (c_i - q)_+, \quad \widehat{D} - S = (\widehat{D} - D) + (D - S).$$

The bar $E_{\max} B_t(a)/(1-\alpha)$ covers only $\widehat{D} - D$ at fixed q/encoding, not population sampling or regime-label uncertainty. $D - S$ combines binning and finite-sample threshold/tail conventions: `sample_cvar` averages observations at/above `numpy.quantile`'s threshold, not necessarily the excess formula with divisor $1 - \alpha$; q need not be the histogram quantile. Hence $D - S$ is not pure quantum error, nor D exact histogram CVaR. **Legacy H is deliberately unchanged: a comparator mismatch**, using width $(\text{hi} - \text{lo})/(2^n - 1)$ versus histogram width $(\text{hi} - \text{lo})/2^n$. Gaussian selected-tail mass: 0.06835, not 0.05; $H = 1.9303992869$, $S = 2.0606125246$. Because H ignores amplitude output, accurate tail-probability estimation cannot repair it (Figure 2). `panel_cvar.py`: 3,090,082 firm-months, 25,048 identifiers, 618 months (July 1973 through December 2024); equal-weighted size/book-to-market quintiles, NYSE breakpoints, at least ten names/month, 221 high-/385 low-volatility months each. Delisting returns are missing; these descriptive conditional samples give no out-of-sample guarantee. At $t = 1024$, all 20 medians meet bars against D, but only seven sample S values occupy those AE-centered intervals: no population coverage claim.

Portfolio / regime	Sample S	Legacy H	Excess D	AE median $\widehat{D}$	AE bar
Size Q1 / high	17.00793	16.40728	17.72510	17.68646	0.27221
Size Q1 / low	9.34103	8.44610	9.40425	9.44051	0.12100
Book/market Q5 / high	17.10908	14.52532	17.57854	17.48918	0.29696

Source: `data/panel_cvar.json`; $t = 1024$, percent monthly loss, bars in percentage points.

Figure 3 (`figures/panel_cvar.png`): `data/panel_cvar.json`, July 1973-December 2024, regenerated 23 October 2026; equal-weighted S, all quintiles/saved regimes, no inferred confidence intervals.

`market_check.py` uses loss = −mktrf (stored monthly market excess return), with 634 labeled months: 223 high, 411 low. These windows differ from the quintile windows; without a population closed form, the same bar covers only fixed D.

Regime	Months	Sample S	Legacy H	Excess D	AE median	AE bar
All labeled	634	10.36406	10.04906	10.30354	10.30477	0.17646
High volatility	223	13.14417	13.16031	13.40383	13.37826	0.18181
Low volatility	411	7.56905	7.40225	7.62960	7.66911	0.09236

Source: `data/market_regimes.json`; $t = 1024$, percent monthly excess-return loss, bars in percentage points.

High-volatility target mismatch $D - S \simeq 0.25966$ points versus median error -0.02557 points: both high/low sample values fall outside AE-centered bars although AE medians meet bars against D. The AE summary covers 17,468 **firm-regime pairs**, not unique firms, with at least 60 months/regime, at $t = 256$: every median inside its bar; median bar 0.01072510 decimal monthly loss = 1.07251 percentage points, not empirical single-shot coverage. Big Red 200 job **8688102**, 22 October 2026 (`BIGRED_RUN.md`), reran `panel_cvar.py --firm-ae` on AMD EPYC: `.out` records cluster path/matching results, `.err` is empty. Big Red reproduced the Mac on a second architecture at reporting precision (maximum absolute JSON difference 1.33×10^{-15}), not literal "last digit" identity. Provenance is the run record, not stale `machine: local Mac`. It did not make the noise more real or the math faster.

Could an Evans-function/winding-number viewpoint organize this error bar by identifying the certified object and whether perturbations preserve its conclusion, as in a Rouche argument? Added value beyond the decomposition is unknown: a question, not a claimed connection or implemented Evans function.

Verified 23 October 2026: 13 numerical/five Qiskit tests passed; redirected `scale_sweep.py` matched saved JSON exactly; `plot_scale.py` regenerated both figures, `plot_panel.py` the sole addition; SHA-256 checked protected data/tests/logs; no panel regeneration. Sources: `NOTE.md`, `MEASUREMENTS.md`, three application/sweep JSONs, `problems.py`, `estimation.py`, `run.py`, Big Red record.